



Department of Physical Therapy

PTH 6122: Dynamic Systems I Movement & Adaptation

Fall 2026

T/Th 8:00 am - 9:15 am, 114 SRH

Course Description

Dynamic Systems I covers movement and adaptation and introduces a system understanding of adaptation and defines movement as a complex system. Movement emerges from physics and biology to develop the clinically useful concept of neuromechanics and motor control, which establish movement as dynamic, complex and adaptive.

Course Information

Credit Hours: 3

Semester Clock Hours: 37.5

Weekly Clock Hours: 2.5

Instructors:

Dr. Eydie Kendall

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Campus Office: 218 Samuel Read Hall

Office hours: T 10-11am; Th 11am-noon; F 12:30-1:30pm or by appt.

Bookings Page: Both in-person and Teams appointments available, including some evenings and weekends. [Book time with Eydie Kendall](#)

Pre Requisites

Clinical & Functional Anatomy, Clinical Physiology, Ethics & Value Systems, Patient / Client Management I, Integrated Clinical I

Co Requisites

Neuroscience, Clinical Inquiry I: Causality & Inference, Patient / Client Management II, Integrated Clinical II

Student Learning Outcomes (Course Objectives):

MODULE 1

1. Understand how identification of disease-associated genetic variations facilitates development of prevention, diagnosis and treatment options. (7A-Genetics)
2. Understand the importance of family history in assessing predisposition to disease in general. (7A-Genetics)
3. Understand the interaction of genetic, environmental and behavioral factors in predisposition to diseases in general including the onset of disease, response to treatment and maintenance of health. (7A-Genetics, Body Systems Interactions)
4. Understand the difference between clinical diagnosis of disease and identification of genetic predisposition to disease (genetic variation is not strictly correlated with disease manifestation). (7A-Genetics)
5. Use information technology to obtain credible, current information about genetics. (7A-Genetics)
6. Summarize the underlying physiological process, principles and the time sequence for neurodevelopment. (7A-Neuroscience, Body System Interactions)
7. Differentiate the sequelae of impaired development during various critical periods of sensorimotor systems. (7A-Neuroscience)
8. Integrate knowledge of maturation of the nervous system with knowledge about automatic and volitional movements. (7A-Neuroscience, Motor Control and Motor Learning)

MODULE 2

9. Explain neuromechanics and motor control as the dynamic integration of mechanics, anatomy, physiology and neuroscience. (7A-Neuroscience, Biomechanics)
10. Compare and contrast current theories of motor control and motor learning (7A-Motor Control and Motor learning)
11. Describe the neuromechanics of human movement and the underlying factors that influence velocity, force, and components of skeletal muscle contraction. (7A-Biomechanics, Motor Control and Motor Learning)
12. Evaluate the neuromechanics and motor control of joint function, force production, static and dynamic posture, activities of daily living and gait. (7A-Neuroscience, Motor Control and Motor Learning, Biomechanics)
13. Describe mechanisms of neural plasticity and their impact on development, recovery of function, and learning. (7A-Neuroscience)
14. Describe factors that influence neural plasticity (e.g., motivation, repetition, age, culture, socioeconomic status [SES], nutrition, and comorbidities). (7A-Neuroscience, Motor Control and Motor Learning, Pain and pain experiences, Psychosocial Aspects of Health and Disability)

Additional Description

Dynamic Systems I covers movement and adaptation. The course expands upon the previous foundation in systems theory considers human movement as a complex system. As a complex system, movement emerges from physics and biology (mechanics, anatomy, physiology and neuroscience) to develop the clinically useful concept of neuromechanics and motor control. Neuromechanics and motor control establish movement as dynamic, complex and adaptive in short time scales as adaptations occur to refine and develop appropriate movement patterns. Longer time scale adaptations (across the life span) are considered including motor learning (including neuromotor development), neural plasticity and neuromuscular adaptations to stress, trauma and aging.

Description of teaching methods and learning experiences

- Synchronous and asynchronous lectures, group discussions, application activities, case presentations.

Assignments and Activities

Quizzes	30%
Exams	40%
Assignments	30%

Student Evaluation/Grading

A	94.0 – 100
A-	90.0 – 93.9
B+	87.0 – 89.9
B	83.0 – 86.9
B-	80.0 – 82.9
C+	77.0 – 79.9
C	73.0 – 76.9
F	below 73.0

Students are reminded that continuation in the program necessitates achieving an overall grade point average of 3.0 (B) or better. Grades below 2.0 (C) are not acceptable or counted toward the DPT degree. No more than 9 credits below 3.0 may be counted toward the DPT degree (grades of B-, C+, or C). Make-up examinations will be given only in specific cases of absence (documented absences). Circumstances that may warrant an absence should be discussed with the instructor before the scheduled time of the exam. Failure to notify the instructor as stated may result in a zero for the exam.

Required Materials:

1. Effgen S. Meeting the physical therapy needs of children. (AccessPhysiotherapy, F.A.Davis collection)
2. Various sections from PhysioU
3. Additional readings and materials as assigned (see Canvas).

Doctor of Physical Therapy Student Handbook: Students are reminded that more detailed information on program policies, including the appropriate and ethical use of Artificial Intelligence (AI), can be found in the Plymouth State University Doctor of Physical Therapy Student Handbook.

Professional Conduct: Students are reminded of the program expectation that they follow Professional Conduct, see the Student Handbook, Professional Conduct and Reasonable Professional Expectations for the process of handling violations, remediation and consequences.

Cell Phones: Cell phone use (including texting) is prohibited in class. Also, please respect your classmates by turning your phone to vibrate prior to class start time. Do not turn phones off as the University uses texts to relate emergency information.

Graduate Academic Policies: The DPT Program is a selective admissions cohort-structured program. Some Graduate Academic Policies may not be relevant to the program. Students should refer to the Doctor of Physical Therapy Program Handbook for specific information about Scheduling, Capstones, and Internships. The sections on Grading System and Academic Standing Policy do apply to DPT students: <https://coursecatalog.plymouth.edu/university-policies-procedures/graduate-academic-policies/>

Fair Grading Policy: Fair and equitable grading reflects values to which all members of the Plymouth State community commit themselves. Grades are used to assess the relative extent to which students achieve course objectives in all for-credit courses at PSU. Academic freedom allows instructors to determine course objectives within the bounds of established curricula, and the means by which a student's mastery of those objectives will be evaluated; and evaluate the quality of work on individual exams or assignments. Information can be found at <https://coursecatalog.plymouth.edu/university-policies-procedures/> under Fair Grading.

Student Support: Student Support Foundation (SSF) provides short-term emergency financial assistance and long-term student support. For more information, see: <https://campus.plymouth.edu/student-support-foundation/>
Student Support Foundation also runs a food pantry, located in Belknap Hall. To learn more about SSF or access the food pantry, either via open hours or a private appointment, contact the SSF advisor, at psu-ssf@plymouth.edu.

Academic Dishonesty: Cheating and plagiarism are very serious forms of academic dishonesty. Any use of unauthorized assistance on exams, papers, homework assignments, or other course work constitutes cheating. Knowingly providing assistance during an exam or allowing other students to copy one's work is also a serious form of academic dishonesty. Plagiarism consists of submitting written work that has been developed wholly or partially by someone else. Submitting written work in which the ideas of others have been duplicated or even

paraphrased without proper reference to the author is also a form of plagiarism. Also considered plagiarism is the acquisition of term papers or other assignments from another source and subsequent presentation of these materials as the student's own work. In addition, students may not use papers in more than one course without the permission of both instructors.

Any student suspected of academic dishonesty will be reported to the Program Director immediately. A full description of academic integrity, violations and procedures when a violation to academic integrity in a course is suspected can be found at: <https://coursecatalog.plymouth.edu/university-policies-procedures/> under Academic Integrity.

Attendance Policy: Attendance is expected to all class sessions. Students are ultimately responsible for their attendance, however faculty will assess students Professional Expectations based, in part, on their attendance (See Student Evaluation/Grading). Students who miss class for any reason are responsible for the information and work missed. Students should speak with the professor before missing the scheduled class when able.

Campus Accessibility Policy: Plymouth State University is committed to providing students with documented disabilities equal access to all university programs and facilities. If you think you have a disability requiring accommodations, you should immediately contact Campus Accessibility Services (CAS) in Speare210 (535-3300) to determine whether you are eligible for such accommodations. Academic accommodations will only be considered for students who have registered with CAS. If you have a Letter of Accommodation for this course from CAS, please provide the instructor with that information privately so that you and the instructor can review those accommodations: <https://campus.plymouth.edu/accessibility-services/>

LibGuide for Physical Therapy:

The Libguide for Physical Therapy is an information sharing platform created through a collaboration between the library and the DPT faculty that presents relevant resources to support students in the Physical Therapy program. Examples of information you will find include tutorials to better understand citing sources in APA and AMA style, resources on how to avoid plagiarism, quick access to relevant databases and interlibrary loan as well as a contact link to the programs' assigned librarian. To view the DPT Program Libguide go to <https://library.plymouth.edu/physical-therapy>.

Course Schedule (tentative)				
Week	Week of	Topics	Assignments	Exams/ Quizzes
1	Sept 2 - 3	Dynamic Systems as a concept – molecules to humans and everywhere in between		
2	Sept 8 - 10	Genetic variations and adaptations		
3	Sept 15 - 17	Interconnection between genome and environment, epigenetics and the relationship between family history and predisposition to disease and dysfunction		Q1
4	Sept 22 - 24	Underlying processes, principles and the time sequence for neurodevelopment		
5	Sept 29 – Oct 1	Sequelae of impaired development during various critical periods of sensorimotor systems		Q2
6	Oct 6 - 8	Motor systems development as a dynamic phenomenon and how this relates to PT		
7	Oct 13 - 15	Cont. & Review		Exam 1 (Th)
8	Oct 20 - 22	TBD		
9	Oct 27 - 29			
10	Nov 3 - 5			Q3
11	Nov 10 - 12			
12	Nov 17 - 19			
13	Nov 24			Q4
14	Dec 1 - 3			
15	Dec 8 - 10			
16		Finals week – date		Exam 2 TBD